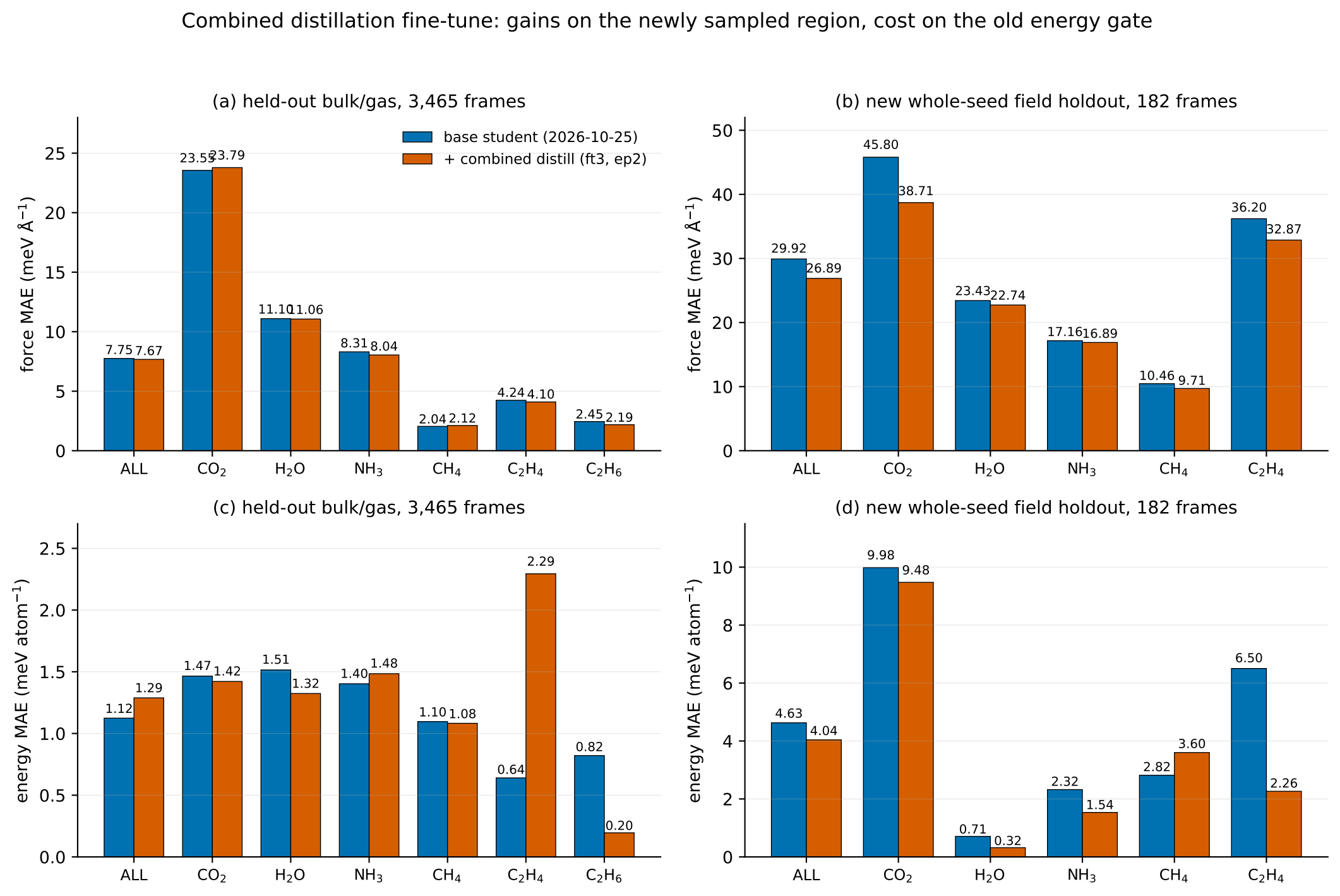
Combined distillation fine-tune: gains on the newly sampled region, cost on the old energy gate
(a) held-out bulk/gas, 3,465 frames
base student (2026-10-25)
+ combined distill (ft3, ep2)
force MAE (meV Å^{-1})
7.75 7.67
23.55 23.79
11.10 11.06
8.31 8.04
2.04 2.12
4.24 4.10
2.45 2.19
ALL
CO_2
H_2O
NH_3
CH_4
C_2H_4
C_2H_6
(b) new whole-seed field holdout, 182 frames
force MAE (meV Å^{-1})
29.92 26.89
45.80 38.71
23.43 22.74
17.16 16.89
10.46 9.71
36.20 32.87
ALL
CO_2
H_2O
NH_3
CH_4
C_2H_4
(c) held-out bulk/gas, 3,465 frames
energy MAE (meV atom^{-1})
1.12 1.29
1.47 1.42
1.51 1.32
1.40 1.48
1.10 1.08
0.64 2.29
0.82 0.20
ALL
CO_2
H_2O
NH_3
CH_4
C_2H_4
C_2H_6
(d) new whole-seed field holdout, 182 frames
energy MAE (meV atom^{-1})
4.63 4.04
9.98 9.48
0.71 0.32
2.32 1.54
2.82 3.60
6.50 2.26
ALL
CO_2
H_2O
NH_3
CH_4
C_2H_4